

1

COMP

Ion distribution and electrochemical gradients

You should be able to: State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

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2

Nernst equation

COMP

You should be able to: Calculate the equilibrium potential for an ion with the Nernst equation and explain what the sign of the result means for the direction that ion will move.

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3

Resting membrane potential

COMP

You should be able to: Explain why the resting membrane potential sits near the potassium equilibrium potential and predict how it shifts when membrane permeability to potassium or sodium changes.

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4

Ion channel gating

COMP

You should be able to: Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

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5

COMP

Depolarization and hyperpolarization

You should be able to: Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

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6

COMP

Membrane potential simulation

You should be able to: Manipulate extracellular potassium and sodium in a simulation and record the resulting change in resting membrane potential against the Nernst prediction.

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7

COMP

Neuron structural and functional classes

You should be able to: Classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

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Functional regions of a neuron

You should be able to: Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

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Glial cell functions

You should be able to: Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

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10

COMP

Myelin and its loss

You should be able to: Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

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COMP

Axonal transport

You should be able to: Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

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COMP

Graded potentials

You should be able to: Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance.

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COMP

Action potential phases

You should be able to: Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

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COMP

Threshold and all or none

You should be able to: Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

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Coding of stimulus intensity

You should be able to: Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

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Refractory periods

You should be able to: Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

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COMP

Conduction velocity

You should be able to: Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

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COMP

Effects of altered extracellular ions

You should be able to: Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

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Action potential simulation

You should be able to: Run a neuron simulation with sodium and potassium channel blockers and interpret the resulting change in the action potential tracing.

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Nerve conduction measurement

You should be able to: Measure conduction velocity from a recorded compound action potential and account for the difference between the fastest and slowest fibers.

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COMP

Sequence at a chemical synapse

You should be able to: Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

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COMP

Neurotransmitter classes

You should be able to: Match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

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Neurotransmitter removal

You should be able to: Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

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COMP

Excitatory and inhibitory postsynaptic potentials

You should be able to: Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change.

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Summation and integration

You should be able to: Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

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Presynaptic modulation

You should be able to: Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

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Synaptic plasticity

You should be able to: Explain long term potentiation as a mechanism of learning and identify the roles of repeated stimulation and receptor insertion.

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Electrical synapses

You should be able to: Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

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COMP

Synapse simulation

You should be able to: Alter neurotransmitter release and receptor availability in a synapse simulation and interpret the resulting postsynaptic recording.

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